Intelligent Document Assistant

SYSTEM EVALUATION & QUERY BLUEPRINT

> This blueprint contains structured evaluation questions designed to test your local Retrieval-Augmented Generation (RAG) pipeline running via Ollama. Use these prompt archetypes to verify retrieval accuracy, context boundary controls, and structural processing capabilities.

1. General Context Extraction (Verify Baseline Retrieval)

Goal: Evaluate if the retriever captures the core thesis.

- **"What are the main limitations and challenges outlined in this document regarding real-time systems?"**
  Intent: Verifies that top-K chunks match your document's primary subject matter.
- **"Summarize the key takeaways from the introductory section in 3 bullet points."**
  Intent: Tests chunk sequential sorting and structural coherence.

2. Specific Fact Extraction (Verify High Precision)

Goal: Isolate explicit metrics, data points, or definitions.

- **"Does the document mention any specific percentages, dates, or performance figures? List them."**
  Intent: Forces the LLM to scan granular tokens inside chunks rather than speaking broadly.
- **"Provide the direct definition or explanation given for the primary problem statement."**
  Intent: Confirms that exact keyword extraction functions correctly across chunk boundaries.

3. Edge-Case / Context Boundary Testing (Verify Safety & Guardrails)

Goal: Confirm system prevents hallucination when outside knowledge is targeted.

- **"Based *only* on the text, what is the step-by-step implementation guide to build this system?"**
  *Intent: Since your document focuses on *challenges* rather than implementation, the model should confidently respond with "I do not know" or clarify missing text.*
- **"Explain how to solve the limitations mentioned using external architectural solutions."**
  Intent: Tests if your prompt template effectively bounds the LLM to your vector space.

4. Comparative & Synthesis Testing

Goal: Evaluate context compilation across distinct chapters.

- **"Compare the initial problems stated in the beginning with the conclusions drawn at the end."**
  Intent: Assesses your chunk retrieval overlap and whether system memory context windows are deep enough.

Engineering Tip for Local Environments:

If processing responses on CPU feels slow, adjust your Streamlit slider parameters down slightly (e.g., **Number of chunks (K) = 3** instead of 5, and **Max New Tokens = 256 or 512**). This minimizes context parsing overload on your system memory and improves generation speeds.